

INDUSTRIAL WATER & WASTEWATER ENGINEERING SERIES

OTHER FILTRATION SYSTEMS

(Multimedia, Carbon, Cartridge, UF/MF, RO) — Detailed Engineering Manual, Expanded Edition

DETAILED ENGINEERING & OPERATIONS MANUAL — EXPANDED EDITION

Water Chemistry Deep-Dives · Process Flow Diagrams · Design, O&M and Project Reference

Industrial Water & Wastewater Engineering Series · Reference Document for Design, O&M and Project Teams

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1. System Profiling & Overview

Filtration systems form the backbone of pretreatment across every water treatment plant covered in this manual series – ETP, STP, cooling towers, DM/softener plants and ZLD trains all rely on one or more of the filtration technologies profiled here. This manual consolidates the design, operation and maintenance knowledge for the filtration systems not covered as standalone plants: multimedia/pressure sand filters, activated carbon filters, cartridge/bag filters, ultrafiltration (UF), microfiltration (MF), and reverse osmosis (RO).

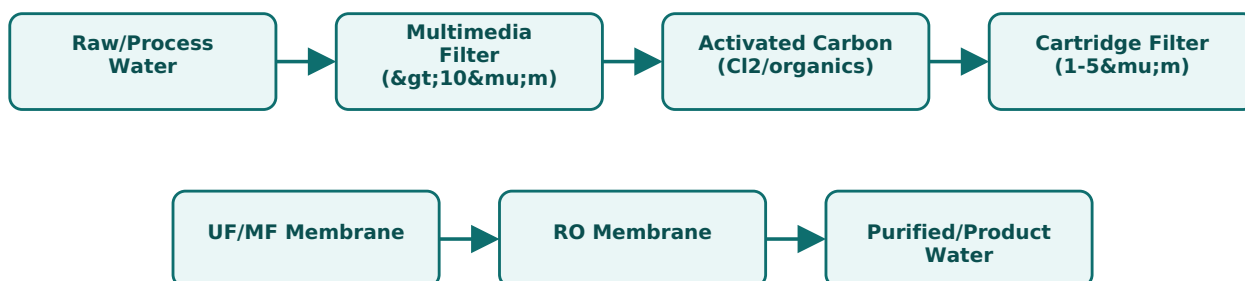


Fig. 1 — Generic Filtration Cascade (coarse to fine): each stage protects the next

1.1 Role in the Overall Treatment Scheme

- Protect downstream ion-exchange resins, membranes and biological systems from particulate fouling.
- Remove specific contaminants (turbidity, color, chlorine, organics, dissolved salts) as a final product-quality step.
- Enable water reuse/recycling by polishing treated effluent to reuse-grade quality.
- Serve as a standalone potable/process water treatment step where raw water source is a borewell, river, or municipal supply of variable quality.

2. Filtration Technology Types & Selection Basis

Technology	Removal Range	Typical Application
Multimedia / Pressure Sand Filter (MMF/PSF)	Suspended solids/turbidity >10-25 micron	First-stage clarification ahead of any downstream process
Activated Carbon Filter (ACF)	Chlorine, organics, color, odor, taste	Dechlorination before RO/resin, odor/color polishing
Cartridge/Bag Filter	1-25 micron fine particulate	Final polishing before RO/UF membranes or dosing skids
Microfiltration (MF)	0.1-10 micron (bacteria, large colloids)	Pretreatment ahead of RO, clarified water polishing
Ultrafiltration (UF)	0.01-0.1 micron (viruses, colloids, macromolecules)	RO pretreatment, tertiary effluent polishing for reuse, dairy/pharma process separation
Reverse Osmosis (RO)	Dissolved ions/salts, 90-99% TDS rejection	Desalination, high-purity water production, ZLD membrane stage
Nanofiltration (NF)	Divalent ions, selective monovalent passage	Softening without full desalination, selective ion separation
Iron Removal Filter	Dissolved/precipitated iron & manganese	Borewell water pretreatment

3. Process Design Basis (each technology)

- **Multimedia Filter:** Graded media (anthracite/sand/garnet) beds designed at 10-15 m/hr service flow rate; backwash at 35-45 m/hr for adequate bed expansion (40-50%).
- **Activated Carbon Filter:** Empty Bed Contact Time (EBCT) of 8-15 minutes for effective chlorine/organics adsorption; service flow 8-12 m/hr; periodic steam/hot water sanitization for biological growth control.
- **Cartridge/Bag Filter:** Sized on differential pressure life (typically replaced at 1.0-1.5 bar dP or per manufacturer's recommended interval), micron rating selected one step finer than the most sensitive downstream equipment.
- **UF/MF Membranes:** Hollow-fiber configuration, operated in dead-end or cross-flow mode with periodic backwash (15-60

min cycles) and chemically enhanced backwash (CEB); flux typically 40-80 LMH depending on feed water quality.

- **RO Membranes:** Spiral-wound thin-film composite elements, staged in a pressure vessel array (e.g., 2:1 or 3:2 array) to maintain concentrate-side velocity; recovery 60-85% for brackish water, limited by scaling potential (LSI/S&DI) and antiscalant program.

4. Pretreatment Requirements

Downstream Technology	Minimum Upstream Pretreatment Needed
Activated Carbon Filter	Multimedia/sand filtration to <10 NTU
Cartridge Filter	Multimedia filtration recommended to extend cartridge life
UF/MF Membranes	Multimedia filtration + cartridge filter (50-100 micron) to prevent fiber damage
RO Membranes	UF/MF or multimedia + cartridge (5 micron), antiscalant dosing, SDI <3-5, dechlorination if thin-film composite membrane used
Iron Removal Filter	Aeration/oxidation stage ahead of filter media to convert soluble to filterable iron

5. Backwash, Cleaning & Membrane Regeneration

- **Media Filter Backwash:** Triggered by time, differential pressure, or turbidity breakthrough; sequence typically air-scour (if fitted) → backwash → rinse to service.
- **Carbon Filter Reactivation:** Exhausted carbon is either replaced or sent for off-site thermal reactivation once adsorption capacity (measured via iodine number or breakthrough testing) is depleted.
- **UF/MF Cleaning:** Regular hydraulic backwash supplemented by periodic Clean-In-Place (CIP) using acid (for scale/metal oxide fouling) and alkaline/oxidant cleaning (for organic/biological fouling).
- **RO Cleaning:** CIP with low-pH cleaner (for scale) and high-pH cleaner (for organic/biofouling) triggered when normalized permeate flow drops 10-15% or normalized salt rejection drops or differential pressure rises 15% above baseline.

6. Membrane & Filtration Chemistry Deep-Dive

6.1 Activated Carbon Adsorption & Dechlorination Reaction

Chlorine reduction on carbon surface (catalytic): $C^* + HOCl \rightarrow C^*O + H^+ + Cl^-$
Organic adsorption: governed by the Freundlich isotherm $q = K \times C^{1/n}$

Unlike physical adsorption of organics (reversible, capacity-limited, described by the Freundlich isotherm), chlorine removal on activated carbon is a catalytic redox reaction that consumes the carbon surface (oxidizes it) over time – this is why chlorine capacity depletes distinctly from organic-adsorption capacity, and why EBCT (contact time) rather than bed mass alone governs reliable dechlorination performance as the bed ages.

6.2 Membrane Rejection & the Solution-Diffusion Model (RO)

Water flux: $J_w = A \times (\Delta P - \Delta \pi)$
Salt flux: $J_s = B \times \Delta C$
where A = membrane water permeability, B = salt permeability, ΔP = applied pressure, $\Delta \pi$ = osmotic pressure difference

RO membranes reject dissolved ions by the solution-diffusion mechanism: water dissolves into and diffuses through the dense polyamide layer far faster than hydrated ions, which are excluded largely by size and charge. This is why RO rejection is a function of applied net driving pressure ($\Delta P - \Delta \pi$) – as recovery rises, concentrate-side osmotic pressure rises and net driving pressure falls, which is the physical reason recovery cannot be pushed indefinitely without adding pressure/stages.

6.3 Fouling Index Metrics: SDI & MFI

Silt Density Index (SDI) = $(1 - t_1/t_2) / T \times 100$
where t_1 = initial 500mL filtration time, t_2 = final 500mL filtration time after interval T (min) through a 0.45µm membrane at 30 psi

SDI < 3 → low fouling risk for RO
SDI 3-5 → moderate, monitor closely
SDI > 5 → high fouling risk, improve pretreatment before RO

SDI is the standard field test for RO feed water particulate/colloidal fouling potential and is required before commissioning any RO system; a rising SDI trend over time (even while turbidity looks stable) is often the earliest warning of upstream media filter or UF degradation.

6.4 CEB & CIP Chemistry for UF/MF

Chemically Enhanced Backwash (CEB) uses low-concentration NaOCl or acid dosed into the backwash water to control biofilm/organic fouling between full CIP cycles. Full CIP alternates a low-pH cleaner (citric acid or HCl, targeting metal oxide/scale fouling) with a high-pH cleaner (NaOH + NaOCl or surfactant, targeting organic/biological fouling) – running cleaners in the wrong sequence, or at the wrong temperature/concentration for the specific membrane polymer, can permanently reduce membrane flux, which is why OEM cleaning protocols must be followed exactly (Section 7).

7. Common Problems & Engineering Solutions

Problem	Root Cause	Engineering Solution
Rapid pressure drop across sand filter	Media fouling, insufficient backwash frequency	Increase backwash frequency, inspect media for mud-ball formation
Carbon filter losing chlorine removal efficiency	Carbon exhaustion, channeling	Test breakthrough, replace/reactivate carbon per schedule
Frequent premature cartridge filter clogging	Inadequate upstream filtration, biological growth	Add/upgrade upstream multimedia filtration, review CIP/sanitization frequency
UF membrane flux decline	Fouling (colloidal, organic, biological)	Increase backwash/CEB frequency, review feed water pretreatment
RO membrane scaling	Exceeding solubility limit (LSI/S&DI), inadequate	Reduce recovery, verify antiscalant dose, improve

	antiscalant	softening/pretreatment
RO membrane biofouling	Inadequate disinfection/biocide control, warm feed water	Implement periodic biocide/CIP program, review feed water microbiological quality
Declining salt rejection (RO)	Membrane degradation from oxidant exposure or age	Verify residual chlorine is nil at RO inlet, plan membrane replacement
UF fiber breakage	Excessive backwash pressure, particulate/abrasive damage, air-scour over-pressure	Review backwash pressure settings, inspect for upstream abrasive solids

8. Equipment & Media Service Life

Equipment/Media	Typical Service Life	Key Life-limiting Factors
Multimedia Filter Media	3-5 years	Media attrition, mud-ball fouling, loss during backwash
Activated Carbon	1-3 years (service dependent)	Adsorption exhaustion, biological fouling
Cartridge/Bag Filter Elements	Weeks to months (consumable)	Differential pressure buildup, micron rating loading
UF/MF Membrane Modules	5-7 years	Fouling, fiber breakage, chemical cleaning degradation
RO Membrane Elements	3-5 years	Scaling, fouling, oxidant damage, compaction
FRP/SS Pressure Vessels	15-20 years	Corrosion, lining wear
Backwash/Feed Pumps	8-10 years (seals/bearings 2-3 yrs)	Cavitation, seal wear
Instrumentation (turbidity/SDI/pressure)	3-5 years	Sensor fouling, calibration drift

9. Preventive Service & Maintenance Schedule

Frequency	Activity
Daily	Monitor differential pressure across all filters; check backwash cycle completion; log RO permeate flow/conductivity/rejection
Weekly	Verify cartridge filter dP trend; inspect UF/MF for visible fiber damage; test carbon filter chlorine breakthrough
Monthly	Normalize RO performance data (flux/rejection/dP); inspect media filter bed for channeling; SDI test on RO feed
Quarterly	CIP of UF/RO per normalized performance trend; carbon iodine number test/replacement planning; media filter bed depth check
Annual	Complete media replacement planning; membrane replacement/autopsy for root-cause fouling analysis; full instrumentation calibration

10. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
High turbidity in filtered water	Media fouling/channeling, broken underdrain	Backwash, inspect underdrain, consider media replacement
Chlorine smell/taste persists post-carbon filter	Carbon exhausted, channeling, insufficient EBCT	Replace/reactivate carbon, verify flow rate against design EBCT
RO permeate conductivity rising	Membrane scaling/fouling, O-ring seal failure, membrane degradation	Perform CIP, inspect for telescoping/seal damage, review antiscalant program
RO differential pressure rising rapidly	Fouling/scaling on lead elements	CIP immediately, review upstream pretreatment
UF integrity test failure	Fiber breakage, O-ring/potting failure	Locate and isolate damaged fiber/module, repair or replace module
Cartridge filter dP rising within days of install	Undersized micron rating, upstream filtration bypass/failure	Verify upstream filter operation, review micron rating selection

11. Project Execution & Engineering Workflow

Phase	Key Deliverables / Activities
Water Quality Study	Raw/feed water characterization, define target product water quality and flow requirement.

Basic Engineering	Technology train selection (which filtration stages needed), PFD, equipment sizing.
Detailed Engineering	P&ID, vessel/skid GA drawings, control philosophy (backwash sequencing, CIP), electrical design.
Procurement	Filter vessels/media, membrane modules, skids, dosing systems, instrumentation.
Civil Works	Foundations for vessels/skids, plant room layout, drainage for backwash/CIP waste.
Erection & MEI	Vessel/skid installation, membrane loading, piping, cabling, control panel wiring.
Pre-commissioning	Hydro test, membrane wet-out/integrity test, backwash sequence dry-run.
Commissioning	Service flow trials, backwash/CIP cycle verification, baseline performance normalization.
Performance Test	Verify product water quality and recovery/flux against guaranteed values at rated flow.
Handover & O&M	O&M manual, baseline performance records, spares list, operator training.

12. Standards & Performance Guarantee

- RO system performance commonly referenced against ASTM D4516 (standard practice for standardizing RO performance data) for normalization and comparison across operating conditions.
- UF/MF membrane integrity testing (pressure decay or diffusive air flow test) is standard practice to verify log removal value (LRV) for pathogen/particle removal claims.
- Typical guarantee parameters: product water quality (turbidity/conductivity/SDI), recovery %, specific energy consumption (kWh/m³), and membrane/media replacement interval, verified over a sustained performance test run at rated flow.
- Guarantee test commonly conducted over 72 hours to 7 days with third-party lab verification of inlet/outlet water quality parameters.

Technical note: figures in this manual are general design guidelines. Final technology train selection, media/membrane sizing and operating limits must be established from actual site water quality data.